MPG No. 5.1.12 $2/m$ [Type I, monoclinic] [G tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & G_y & 0 \end{bmatrix}$$

$$G_y = \mathbb{G}_y^{(1)}$$

* Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & G_{xxy} & 0 \\ 0 & G_{yyy} & 0 \\ 0 & G_{zzy} & 0 \\ G_{yzx} & 0 & G_{zzy} \\ 0 & G_{yzx} & 0 \\ G_{xxy} & 0 & G_{yzx} \end{bmatrix}$$

$$G_{xxy} = \mathbb{G}_y^{(1)} - \mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\beta(1)}$$
$$G_{yyy} = 3\mathbb{G}_y^{(1)} + 2\mathbb{G}_y^{\alpha(1)}$$
$$G_{zzy} = \mathbb{G}_y^{(1)} - \mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\beta(1)}$$
$$G_{yzx} = \mathbb{G}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & G_{xxy} & 0 \\ 0 & 0 & 0 \\ 0 & G_{zzy} & 0 \\ G_{yzx} & 0 & -\frac{G_{zzy}}{2} \\ 0 & G_{zxy} & 0 \\ -\frac{G_{xxy}}{2} & 0 & -G_{yzx} - G_{zxy} \end{bmatrix}$$

$$G_{xxy} = 2\mathbb{G}_y^{(2)} - 2\mathbb{Q}_{xz}^{(1)}$$
$$G_{zzy} = 2\mathbb{G}_y^{(2)} + 2\mathbb{Q}_{xz}^{(1)}$$
$$G_{yzx} = 3\mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)}$$
$$\frac{G_{zzy}}{2} = \mathbb{G}_y^{(2)} + \mathbb{Q}_{xz}^{(1)}$$
$$G_{zxy} = -3\mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)}$$
$$\frac{G_{xxy}}{2} = \mathbb{G}_y^{(2)} - \mathbb{Q}_{xz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} G_{yzx} & 0 & G_{yzz} \\ 0 & G_{zxy} & 0 \\ G_{xyx} & 0 & G_{xyz} \end{bmatrix}$$

$$G_{yzx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(2)} + \mathbb{Q}_v^{(2)}$$
$$G_{yzz} = -\mathbb{G}_y^{(3)} + \mathbb{Q}_{xz}^{(2)}$$
$$G_{zxy} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(2)} - \mathbb{Q}_v^{(2)}$$
$$G_{xyx} = \mathbb{G}_y^{(3)} + \mathbb{Q}_{xz}^{(2)}$$
$$G_{xyz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(2)}$$

* Rank 4 tensor (sss). * Rank 4 tensor (ssa). * Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s). * Rank 4 tensor (a). * Rank 4 tensor (t).